

Exercise 4: Table Calculations and Parameters

Three worksheets, three captions. No memo.

Item	Details
Points	100
Data	Exercise 4 starter workbook, or clean/mvu_enrollment_fact.csv (Worksheets 1–2) and clean/mvu_student_dim.csv (Worksheet 3)
Reading	Learning Tableau 2025 (Milligan, 6th ed.), free through UTD Libraries: Ch 6, plus the parameters material in Ch 4 and Ch 11
Walkthrough	Module 4 of the MVU Tableau Walkthrough follows these same steps, with checkpoints. Start with its review page if Modules 1–3 feel shaky.
Submit	One packaged workbook (.twbx) with Worksheets 1–3 and a visible caption on each. No memo.

Worksheet 1: running total, addressing and partitioning

1. Connect mvu_enrollment_fact.csv. college on Rows, term_code on Columns, a record count on Text (the table's Count field, or student_id set to Count, not Count Distinct).

Checkpoint: Total enrolment records across all terms and colleges?

2. Right-click the measure > Add Table Calculation > Running Total. Cycle through Table (across), Table (down) and Pane (down) and watch the numbers.
3. Finish on Specific Dimensions with term_code ticked and college unticked. Add Analysis > Totals > Show Column Grand Totals.

Checkpoint: Final running-total value for the last term, in the grand total row?

Worksheet 2: percent difference and a moving average

1. New worksheet. term_code on Columns, the record count on Rows (a line chart over six terms).
2. Quick Table Calculation > Percent Difference From, set to Previous along term_code. Add a constant reference line at 0.

Checkpoint: Percent difference from 2021SP to 2021FA, to the nearest whole percent?

3. Duplicate the sheet and change the calculation to a Moving Average of 2 prior periods plus the current one.
4. Build the honest version too: filter to fall terms only, then apply percent difference.

Worksheet 3: a parameter, several metrics

1. Create a String parameter, Select Metric, with a list: Headcount, Graduation Rate, Average Aid.
2. Connect mvu_student_dim.csv and create Dynamic Measure with CASE [Select Metric]: COUNTD([student_id]), AVG([is_graduated]), AVG([total_aid_awarded]).
3. Show the parameter control. college on Rows, Dynamic Measure on Columns. Switch metrics.

Checkpoint: With Headcount selected, what is the total across all colleges?

Your three captions

Each worksheet carries a caption: a few sentences answering its question, written while the view is in front of you. Turn it on with Worksheet > Show Caption, then double-click below the view to edit.

Sheet	What your caption must answer
Worksheet 1	Which dimension does the calculation move along, and which does it restart within? Which Compute Using option was meaningless, and why?
Worksheet 2	What causes the $\pm 100\%$ swings? Did the moving average reveal a trend or conceal a pattern? If the periods aren't comparable, which comparison would be valid?
Worksheet 3	The three metrics sit on very different scales. What problem does that create for a reader, and how did you handle it?

Worksheet 2's caption carries 25% on its own. A caption saying the moving average “made the trend clearer” does not earn full marks. That question is the point of the exercise; the other two assess whether you can operate the software.

Before you submit: open each sheet and confirm the caption is showing. A caption toggled off still holds your text but won't be visible when your instructor opens the workbook.

Submit for Exercise 4

- Worksheet 1: running total, scoped through Specific Dimensions rather than Table (across)
- Worksheet 2: percent difference with a zero reference line, plus the moving-average version
- Worksheet 3: a parameter-driven view switching between three metrics
- A visible caption on each
- Packaged workbook (.twbx). No memo.